

Fernando Lopez

872-305-7616 | fernandolpz.dev | flope8@uic.edu | [linkedin.com/in/fernando-lopez-dev](https://www.linkedin.com/in/fernando-lopez-dev) | github.com/FernandoLpz0911

Education

University of Illinois Chicago (UIC) Bachelor of Science in Computer Science	Expected: May 2027
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Technical Skills

Programming Languages: Python, C++, Java, Go, SQL, JavaScript, Dart
Frameworks: LangChain, React, Django, Flutter, Pytest, JUnit5
Deployment Tools & APIs: Vercel, Render, Sanity, Brevo, EmailJS
Development Tools: Docker, Github Actions, Git, Linux, Vim, Firebase, PostgreSQL, SQLite, Figma, Maven, Arduino IDE, Agile
AI Models: WatsonX

Relevant Experience

University of Illinois Chicago (UIC) - Medical Research Assistant	Jan 2026 - Present
- Architected and deployed Voxylex AI utilizing a CI/CD pipeline (GitHub Actions) to automate testing and delivery of a Flutter/Dart mobile application for patient speech analysis. - Overhauled Firebase infrastructure (Firestore/Cloud Storage) via advanced indexing, reducing database read/writes by 98%. - Integrated App Check and Crashlytics telemetry within an MVVM architecture to enforce robust security protocols and centralized real-time error logging.	
Dulce Tentaciones - Freelance Software Engineer	Jun 2025 - Aug 2025
- Directed the full SDLC for a local bakery startup, delivering a responsive React application from requirements to deployment - Engineered a scalable marketing pipeline using Brevo and EmailJS, capable of processing 1,000+ monthly customer orders without manual intervention - Integrated Sanity CMS to enable non-technical content updates, reducing developer maintenance time by 100%	
Compudopt Inc - Program Teacher	Aug 2024 - Feb 2025
- Translated complex programming paradigms into accessible K-8 curriculum, covering foundational logic such as loops, control flow, and variable types to build technical literacy for 20+ students per semester - Directed hands-on engineering workshops utilizing microcontrollers (Arduino) and circuit logic to foster early proficiency in embedded systems and real-world software use cases	

Projects

Predictive ML Fitness Model (Python, XGBoost, TensorFlow, Pandas, NumPy, Scikit-learn)	Apr 2026
- Architected an XGBoost machine learning pipeline to transform raw time-series fitness data into an autonomous decision engine, predicting physiological strength capacity (1RM) and eliminating historical training plateaus. - Evaluated and transitioned model architectures, pivoting from a Multi-Layer Perceptron (MLP) neural network to a Gradient Boosted Tree (XGBoost) ensemble to better handle sparse tabular data and improve prediction stability. - Engineered dynamic time-series features by applying piecewise physiological formulas (Brzycki, Epley, Mayhew) to normalize raw workout logs into robust predictive data points.	
RAG PDF Assistant - Michigan DevFest (6-Hour Hackathon) (Django, React, Docker, WatsonX)	Nov 2025
- Developed a context-aware RAG AI for document analysis using IBM WatsonX and LangChain within a 6-hour hackathon - Engineered a containerized Django/Docker backend to efficiently handle PDF processing and vector embeddings, ensuring environment consistency - Deployed a decoupled architecture via Render (backend) and Vercel (frontend), integrating Firebase for secure authentication and persistent data storage	
Digital Art Watermark Page (Python, Django, React, PostgreSQL, Figma, Javascript, Render, Vercel)	Aug 2025 - Dec 2025
- Engineered a secure, decoupled, full-stack permission system utilizing advanced data structures for digital rights verification - Implemented a Merkle Tree architecture to ensure tamper-proof data integrity and permission verification - Integrated Bloom Filters and Skip Lists to optimize performance, achieving O(1) existence checks and O(log n) search speeds - Designed the PostgreSQL schema and deployed the modularized architecture using Vercel (Frontend) and Render (Backend)	
Cross-Platform Notes App (C++, Dart, Flutter)	Jun 2025
- Architected a cross-platform MVP using Flutter and Dart, delivering a single unified codebase for both iOS and Android deployment - Architected a high-performance local data layer in Dart, achieving a verified throughput of 1.5M+ operations/second via rigorous automated stress testing - Designed a scalable Object-Oriented architecture prioritized for memory efficiency, capable of managing an initial local data volume of 1 GB - Implemented Firebase Authentication and Realtime Database to ensure secure user sessions and instant multi-device synchronization	

Leadership & Activities

Women in Computer Science - Aspire To Inspire Mentor	Sep 2025 - Present
- Led a 4-person agile team to architect and develop a real-time chatting application utilizing Django and React. - Mentored 4 students in full-stack development, guiding the end-to-end deployment of their personal web portfolios. - Drafted technical roadmaps defining the scope for future AWS deployment and cross-platform mobile expansion.	